

Do **not** write solutions on this page.

11. [Maximum mark: 20]

(a) It is given that $w = 1 + i\sqrt{3}$.

(i) Express w in the form $w = re^{i\alpha}$, where $r > 0$ and $-\pi < \alpha \leq \pi$.

(ii) Show that $(1 + i\sqrt{3})^8 + (1 - i\sqrt{3})^8 = -256$. [6]

It is given that $(1 + i \tan \theta)^n \equiv \frac{\cos n\theta + i \sin n\theta}{\cos^n \theta}$ for $\cos \theta \neq 0$, $n \in \mathbb{Z}^+$.

(b) Show that $(1 + i \tan \theta)^4 + (1 - i \tan \theta)^4 \equiv \frac{2 \cos 4\theta}{\cos^4 \theta}$. [3]

(c) Hence, determine the four roots of the equation $(1 + z)^4 + (1 - z)^4 = 0$, giving your answers in the form $z = i \tan\left(\frac{k\pi}{8}\right)$, where $k \in \mathbb{Z}^+$. [4]

(d) By considering the binomial expansion of $(1 + z)^4 + (1 - z)^4$, show that $\tan \frac{\pi}{8} = \sqrt{3 - 2\sqrt{2}}$. [7]

